SAROTA ONDIEKI RAPHAEL

Graduate Telecommunication Engineer

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PROFESSIONAL SUMMARY

Motivated Graduate Telecommunication and Information Engineer with practical hands-on experience gained through multiple industrial attachments in telecommunications, broadcasting, networking, and software development environments. Skilled in RF signal testing fundamentals, fiber optic deployment and testing, structured cabling, IP networking basics, and network performance monitoring. Experienced in supporting telecom infrastructure operations, troubleshooting connectivity issues, and documenting technical processes. Strong analytical ability with programming, and a solid foundation in modern telecommunications and network technologies. Quick learner with strong teamwork, problem-solving, and field-work discipline.

TECHNICAL SKILLS

Mobile Core Networks: 2G, 3G, 4G LTE (RSRP, RSRQ, SINR, throughput, KPIs), SSV, drive testing, TEMS Investigation
Programming: Python,R,STATA, JavaScript, Arduino, C++, Full-Stack Development
Automation: REST API, JSON, Ansible (basics), Git
Tools: Git, Linux Shell scripting, VS Code, Microsoft Office
WAN Routing: Static Routing,IP
Switching: VLANs, Trunking, basic Cisco switch configuration
Fiber Optic Systems: Fusion Splicing, OTDR Testing, Link Loss Testing, FTTH
Network Security: SSH, basic ACLs

PROFESSIONAL EXPERIENCE

Afrotel Engineering Group, Nairobi, Kenya

Drive Test and SSV Intern **Jan 2026 – Present**

- Conduct drive tests and Single Site Verification (SSV) for LTE sites post-integration, ensuring readiness for commercial launch
- Collect and log real-time RF parameters including RSRP, RSRQ, SINR, throughput, and call success rates, achieving data accuracy for optimization team analysis
- Support cluster verification ,identifying and reporting coverage gaps and interference issues for immediate resolution
- Prepare comprehensive drive test logs, route documentation, and site acceptance reports.
- Perform basic post-processing and KPI analysis, contributing to improvement in handover success rates through early issue detection

INDUSTRIAL ATTACHMENTS

Nuricha Technologies, Juja, Kenya **May – August 2024**

Software Engineering Trainee

- Built responsive web applications using React.js and Tailwind CSS, delivering seamless user experiences.
- Integrated RESTful APIs to enable real-time data fetching and updates, significantly improving application interactivity and responsiveness
- Explored and implemented AI-enhanced UI features (chat interfaces, smart suggestions, personalization elements) to improve user engagement and overall platform experience
- Participated in code reviews, UI/UX discussions, and agile development processes, gaining practical exposure to modern front-end workflows

Kenya Broadcasting Corporation (KBC), Nairobi, Kenya **May – Aug 2023**

Broadcast Transmission Trainee

- Supported monitoring and basic troubleshooting of IP and fiber-based studio-to-transmitter links.
- Participated in RF signal path testing**,** helping to locate and resolve frequency interference issues.
- Supported network connectivity for studio and transmission sites, ensuring stable data and audio-visual feeds.
- Gained exposure to satellite uplink and downlink systems, including encoder configuration and link budget monitoring
- Contributed to preventive maintenance activities and documentation of transmission equipment status

Kenya Power Limited Company (KPLC), Thika, Kenya **May – Jul 2022**

Networking and IT Trainee

- Performed fiber optic fusion splicing and OTDR testing to achieve low-loss connections and verify link quality during network installations and maintenance activities
- Assisted in structured cabling works including UTP terminations, patch panel dressing, and cable testing to ensure reliable physical layer connectivity
- Supported basic TCP/IP configurations on network devices and assisted in IP address assignment and basic connectivity verification
- Helped document and label cable routes, patch panels, and network points, significantly improving traceability and future troubleshooting efficiency
- Configured and performed basic troubleshooting of network equipment including routers, switches, and VoIP phones to support internal communication systems
- Contributed to fault identification and resolution efforts, helping maintain reliable voice and data services across departments

EDUCATION

Jomo Kenyatta University of Agriculture & Technology (JKUAT)
BSc. Telecommunication and Information Engineering - Upper Second Honors (2020-2025)

Cardinal Otunga Boys High School

Kenya Certificate of Secondary Education (KCSE): A- (Minus) 2016 – 2019

TECHNICAL PROJECTS

Optic Link Budget Analyzer

- Created a Python-based tool to calculate link losses, splice efficiencies, and overall budget for FTTH deployments.
- Tested on sample fiber routes, identifying attenuation issues and recommending splicing improvements.
- Incorporated data visualization for reports, aiding in preventive maintenance planning.

Intelligent Audience Tracking and Analytics System | Final Year Project 2025

- Designed and developed an intelligent system for real-time audience tracking and behavior analytics in public venues and events
- Integrated computer vision algorithms, sensor fusion, and data analytics to monitor crowd density, movement patterns, and demographic insights
- Built a dashboard for visualization of key metrics (occupancy, dwell time, flow rate) to support venue management and safety decisions
- Implemented data processing pipeline using Python and basic machine learning techniques for actionable insights

VOLUNTARY EXPERIENCE

Jomo Kenyatta Gusii Students Association (JGUSA) – JKUAT **Nov 2023 – Nov 2024**
Secretary General

- Acted as official spokesperson and communications lead.
- Chaired Welfare Committee, managing student support and advocacy programs.
- Liaised with county leadership to coordinate student engagement initiatives.

CERTIFICATIONS

- Cisco Certified Network Associate (CCNA)
- Cisco Certified DevNet Associate (DevNet) – Ongoing Course
- Power Learn Project (PLP) Software Engineering

SOFT SKILLS

Technical Documentation & Accurate Reporting
Clear Verbal and Written Communication
Fast Learning & Adaptability
Field Work Discipline

REFERENCES

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